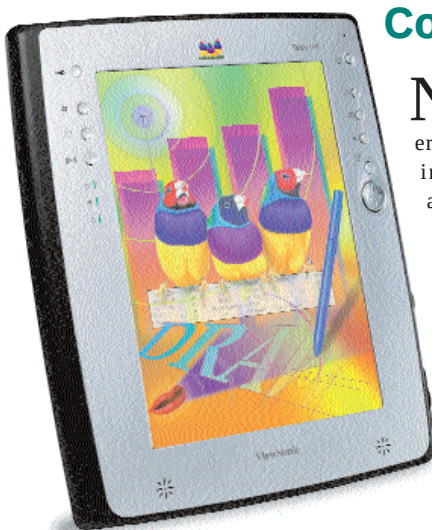
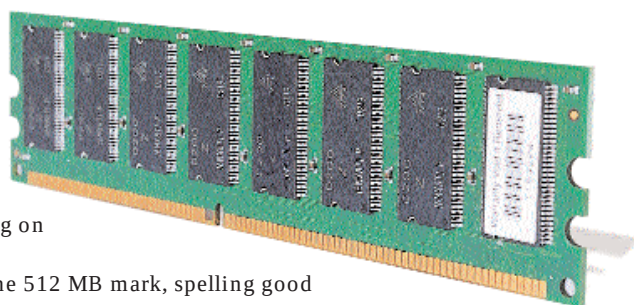


The memory angle

After the processors, probably the single most important component that spells computing performance is the memory. The past year has seen rapid advancements in the memory scene—the RDRAM and the DDR camps have been churning out faster modules with denser capacities.

While both have gained acceptance, the future clearly stands in favour of DDR memory. It has been very visible in the AMD and Intel platforms and its newest iteration, DDR2, is poised to be the memory platform of choice for both Intel and AMD. Making its way to this market is the DDR-III, which yet again promises to provide a worthy complement to the forthcoming processor platforms with bandwidths ranging from 6.4 GBps all the way to a blistering 25.6 GBps—faster than the PC4300 DDR2 memory bandwidth of 4.3 GBps. This year will see DDR-III appearing on motherboards across the Intel and AMD platforms.

On the topic of density, these RAM modules have already touched the 512 MB mark, spelling good things for the server and enterprise segments. However, with future operating systems and application programs demanding greater amounts of memory, greater densities will also mean lower costs per module.



Computing to go

No longer is the desktop the only place to get work done. The newest slew of handheld devices—cellular phones, Tablet PCs, handheld devices and new-age notebook computers—have seen advancements in processing power, battery life and storage capacities, resulting in an array of fairly advanced capabilities. Cellular phones such as the Sony Ericsson T68i and Nokia's Communicator are prime examples of this yet unrealised potential.

This year will see an explosion in devices that let you take your communication, documents and even Internet on the road with you. Devices that incorporate imaging capabilities such as the Sony Ericsson P800 will be widespread in this year. Handheld devices based on the Windows Pocket PC and Palm platform have already demonstrated support for seriously powerful processors with full imaging, audio and video capabilities.

In 2003, also watch out for the emergence of Tablet PCs—the next mobile computing device. Delivering high levels of user-friendliness and functionality, Tablet PCs, along with forthcoming cellular and handheld devices will revolutionise mobile computing.

Display advancements

They've been around for the longest time—CRT monitors are trusty devices that have lasted all through the years, with very little of the underlying technology changing. However, LCD displays, once the devices of the privileged, are now within our budgets due to enhancements in fabrication processes and yields. With all the advantages of smaller footprints and greater brightness and contrast levels, LCD technology is fast catching up on the price and resolution front as well. The year will see the arrival of LCD displays that are affordable, with screen sizes in the 14 and 15-inch range.

One of the display technologies to look out for is Organic LED. Displays based on this technology eliminate the need for a power source for backlighting, spelling significantly lower manufacturing costs along with higher contrast and brightness levels. Its response time will be about 1000 times that of present-day LCDs—no more blurring even when viewing fast-moving images or large displays!

